Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · JavaScript · TypeScript · Java · Object-Oriented Design · CI/CD · Automated Testing · REST APIs · AWS · AI/ML Integration · Generative AI · Scalable Backend · Observability

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PROFESSIONAL SUMMARY

Software engineer with 8 years delivering **clean, scalable, production-grade backend systems** through iterative, scoped delivery across the full software development lifecycle - design, object-oriented coding, automated testing, CI/CD, deployment, and production support. At Entrata: owned **20+ production modules** (Python, PHP, JavaScript, PostgreSQL, AWS S3, RabbitMQ) serving **100K+ monthly active users at 99.5% uptime** across 120+ Agile sprints; optimized 200+ SQL queries (30% performance gain); led Docker microservices redesign (50% scalability improvement, 30% deployment reduction); 300+ code reviews, 5+ engineers mentored. Built **AgentProbe**: open-source **AI/ML evaluation platform** with 5-metric eval engine, **CI-integrated automated testing**, observability dashboard, and multi-provider generative AI routing (OpenAI, Anthropic, Groq, Ollama) - used to evaluate AI components at scale. Built **Leasel**: enterprise AI platform integrating **generative AI with deterministic guardrails** to automate complex compliance workflows. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

WORK EXPERIENCE

George Mason University

Aug 2024 – May 2026

Research Associate / Graduate Teaching Assistant — Python · ML Modeling · Analytics · Curriculum Delivery

- Built multi-source prediction pipelines (Python, R, 190+ countries) — **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; produced Tableau and ggplot2 observability dashboards driving 40% improvement in comprehension for faculty leadership. Delivered technical workshops to **600+ students** covering Python OOP, data structures, and database design.

Entrata — Enterprise PropTech SaaS

Sep 2018 – Aug 2024 · Remote

Senior Software Engineer — Full-Stack · OOP · CI/CD · REST APIs · AWS · Scalable Backend · AI Integration

- **Owned 20+ full-stack production modules end-to-end** (Python, PHP, PostgreSQL, JavaScript, TypeScript, RabbitMQ, AWS S3) across 120+ Agile sprints serving **100K+ MAU at 99.5% uptime** - object-oriented design, schema normalization, multi-tenant data isolation, integration testing, and CI/CD deployment for every release.
- **Led microservices redesign with Docker and CI/CD pipelines** (GitHub Actions) - 50% scalability improvement, 30% deployment time reduction; **optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, partitioning, audit tables, and triggers; built **automated testing and observability tooling** that kept open bugs below 10 for 6+ consecutive months.
- **Integrated AI/ML pipeline** (Python, PyTorch, OpenAI API) replacing manual Excel workflows on 1.6M+ federal records; built **RabbitMQ async processing** with retry logic and idempotency guarantees eliminating double-processing failures; led PHI-compliant AWS S3 migration with Libsodium encryption (100% compliance, zero security incidents); **300+ peer code reviews; mentored 5+ engineers**.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018 · Pune

Associate / Trainee Software Engineer — Full-Stack · OOP · Data Pipelines · AWS S3

- Built object-oriented property management modules (PHP, Python, PostgreSQL, JavaScript); designed SQL import pipelines for 1.6M federal housing records; led AWS S3 cloud migration; participated in sprint planning, code reviews, and performance debugging under senior engineer guidance.

PROJECTS

AgentProbe: AI/ML Evaluation Platform with CI/CD & Observability

AI EVAL · OPEN SOURCE

Python · LangGraph · FastAPI · ChromaDB · OpenAI / Anthropic / Groq / Ollama · React · CI/CD (GitHub Actions) · Docker — github.com/venkatalakshmiparimala/AgentProbe

- **Full-stack AI/ML evaluation platform**: 15-endpoint FastAPI backend (object-oriented, modular), **multi-provider generative AI routing** (OpenAI, Anthropic, Groq, Ollama), 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), **CI-integrated automated testing** gate blocking quality regressions on every commit, and React **observability dashboard** tracking per-run metrics and retrieval drift; any AI component integrates via REST API without code changes. Validated AI output across 1,240+ test cases: pass@1 81.7%, pass@3 88.3%.

Leasel: Enterprise AI Platform — Generative AI + Deterministic Guardrails

AI INTEGRATION

Python · LangGraph (8-agent) · FastAPI · Streamlit · RAG · OCR · SQLite · REST APIs — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Scalable, maintainable enterprise AI platform** built on object-oriented Python with 8-agent orchestration, FastAPI backend, Streamlit engineer console, and RAG over compliance documents; **integrates generative AI with deterministic business-logic guardrails** to enhance software intelligence while enforcing correctness on regulated decisions; CI regression gate, structured logging, and confidence-scored outputs for full observability. Results: 93.6% precision, 0% unsupported-claim rate on 1,240 compliance documents.

AI on FHIR: NLP-Powered Healthcare API Integration

AI INTEGRATION · FULL-STACK

Flask · React · BERT · LLMs · FHIR REST API · Docker · spaCy · JavaScript — github.com/venkatalakshmiparimala/AI_on_FHIR

- Flask + React platform **integrating AI/ML components (BERT, LLMs) with a third-party REST API** to translate natural-language queries into structured API calls; ~60% reduction in manual retrieval time; HIPAA-compliant RBAC; containerized with Docker. Demonstrates end-to-end AI/ML integration across a full-stack production-style system.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

Software systems, ML pipelines, data engineering, AI integration, statistical modeling, observability

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Engineering & Backend: Python (expert), JavaScript, TypeScript, PHP, Java (familiar), SQL (expert — 200+ production optimizations, schema design, execution plan tuning, indexing, partitioning, triggers, audit tables), PL/SQL, R, HTML/CSS; Object-Oriented Programming (OOP), Full Software Development Lifecycle, REST APIs, JSON, Microservices, Docker, CI/CD (GitHub Actions), Automated Testing, Observability, AWS (S3, EC2, RDS, Glue), RabbitMQ, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum

AI/ML Integration & Generative AI: AI/ML component integration, Generative AI tools (OpenAI API, Anthropic API, Groq, Ollama), LangGraph, LangChain, RAG Pipelines, Multi-Agent Orchestration, Prompt Engineering, LLM Evaluation, AI Output Validation, Deterministic Guardrails, FastAPI, ChromaDB, NLP, BERT, spaCy, Hugging Face

ML & Analytics: PyTorch, scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, Statistical Modeling, Tableau, Power BI, ggplot2; 300+ code reviews, 5+ engineers mentored